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ORBITAL CLIMATE FORCING IN MUDFLAT TO MARGINAL LACUSTRINE DEPOSITS IN THE MIOCENE TERUEL BASIN (NORTHEAST SPAIN)

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ABSTRACT: **Upper Miocene mudflat and marginal lacustrine sediments were deposited on the fringes of an alluvial-fan system in the low-gradient, endorheic southern Teruel Basin, northeast Spain. The 147-m-thick Prado succession consists of 55 meter-scale sedimentary cycles, displaying a clastic-rich and a carbonate-rich interval. The clastic-rich parts are weakly pedogenically modified, red to yellow-orange mudstones and are interpreted as deposits of a well-drained mudflat. The carbonate-rich sediments consist of calcretes, marls, and muddy limestones. The calcretes range from dispersed nodules to massive nodular and are interpreted to be of pedogenic or phreatic origin. The marls and muddy limestones are light- to dark-gray and rich in organic matter, shell fragments, and gastropods. They are interpreted as ephemeral pond to marginal lacustrine deposits. A composite sedimentary cycle model is presented summarizing the different facies and their vertical distribution throughout the Prado area. On a 10 m scale, distinct stratigraphic intervals show lithologic trends representing alternating wetter and drier paleoenvironments. Cycle characteristics are laterally consistent, whereby limestone beds pinch out towards the basin margin and thicken towards the central parts of the 10-km-wide and 20-km-long subbasin.**

The high-resolution magnetostratigraphy resolves four normal-polarity and four reversed-polarity intervals that are straightforwardly correlated to chron interval C5n.2n to C5Ar.1r in the ATNTS04 time scale. The resulting age control indicates a duration of 19 to 23 kyr for the meter-scale lithofacies cycles, suggesting control by precession-driven climate change. An astronomical tuning is established by correlating each cycle to the La2004 precession curve using recently recalibrated ages for the polarity reversals as a starting point. The resulting astronomical ages for the reversals in the Prado section are consistent with the astronomically calibrated magnetostratigraphy and cyclostratigraphy of the marine Monte dei Corvi section in Italy. At Prado, three intervals with thicker-than-average cycle thickness most likely reflect obliquity influence during 405 kyr eccentricity minima when precession amplitudes are low. This corroborates the orbital forcing hypothesis of the lithofacies cycles and the astronomical tuning of the Prado succession. The combined magnetostratigraphy and cyclostratigraphy method applied in this study shows the high potential of this method for detecting orbital climate forcing in continental successions.

1. INTRODUCTION

The recognition of astronomically forced sedimentary cyclicity in continental basin fills can considerably enhance both relative and absolute age control (Kent and Olsen 1999; Hinnov 2000), understanding of the sedimentology (Meyers et al. 2001; Weedon 2003), quantification of rates and duration of sedimentary, climatic, and paleontological processes (Fischer and Roberts 1991; Meyers et al. 2001), and possibilities for correlation within and beyond the basin (Van Vugt et al. 1998). The influence of astronomical climate forcing has now been demonstrated and suggested for various types of lacustrine successions with different geological ages (Bradley 1929; Eardley et al. 1973; Olsen 1986; Astin 1990; Fischer and Roberts 1991; Abdul Aziz et al. 2000; Reinhardt and Ricken 2000; Luzón et al. 2002). The detection of astronomical cyclicity requires sufficiently long sediment records (Olsen 1986; Abdul Aziz et al. 2003) or a first-order reliable age control, that can be magnetostratigraphic (Van Vugt et al. 2001; Kruiver et al. 2002; Abdul Aziz et al. 2004) or radiometric dating of ash layers intercalated within the sediment record (Pietras and Carroll 2006; Meyers and Sageman 2007; Machlus et al. 2008). The geomagnetic polarity time scales of the Neogene, and recently the Paleogene, have been considerably improved by astronomically calibrated ages for polarity reversals (Lourens et al. 2004; Pälike et al. 2006; Westerhold et al. 2008). These improved ages allow detection of orbital climate forcing in continental sediment records and construction of astrochronologies in even relatively short sedimentary successions (Van Vugt et al. 1998; Steenbrink et al. 1999; Abdul Aziz et al. 2004; Abels et al. 2009), thus avoiding statistical analyses that are reported to often contain a certain degree of subjectivity (Wilkinson et al. 2003; Meyers and Sageman 2007; Bailey and Smith 2008).

The upper Miocene Cascante section in the southern Teruel Basin (NE Spain) displays very regular, meter-scale cyclicity of alternating red-brown, distal-alluvial-fan mudstones and palustrine to shallow lacustrine limestones. High-resolution magnetostratigraphic age control, substantiated by mammal biostratigraphy, revealed that this meter-scale cyclicity is related to the climatic precession cycle (Abdul Aziz et al. 2004).

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Carbonate microfacies analysis of limestone beds of successive meter-scale cycles revealed the additional presence of larger-scale lake-level variations that are related to the short and long eccentricity cycles. These results allowed construction of an astrochronology for this part of the basin (Abels et al. 2009). Roughly time-equivalent deposits are present in the Prado area, 6.5 kilometers NNW of the Cascante section and located in the same structural segment of the Teruel Basin, but at the opposite basin margin.

The succession at Prado shows different lithological characteristics than the succession in the Cascante area, being dominated by yellow-orange to red-brown mudstones and calcretes from (relatively) well-drained mudflats and marginal lacustrine calcareous mudstones and muddy limestones. In analogy with the Cascante section, the Prado section is characterized by meter-scale, mudstone–limestone cycles over its whole stratigraphic extent, but with a higher variability in sedimentological characteristics, both laterally and vertically. On a larger scale, stratigraphic intervals dominated by limestones or calcareous mudstones alternate with intervals characterized by prevalent mudstone deposits. The rough time equivalence with the Cascante succession is known from formation-scale correlations in the relatively small basin and from preliminary small-mammal biostratigraphic results. A direct, bed-to-bed, lithological correlation to the Cascante section can, however, not be achieved, either physically or by matching lithological characteristics. Therefore, detailed magnetostratigraphic sampling has been carried out to obtain a first-order age control that allows detecting orbital forcing within the Prado section, and, subsequently, constructing an independent astrochronological framework for this part of the Teruel Basin. The vertically and laterally more complex meter-scale cyclicity and dominant long-term changes in the sedimentary successions of the Prado area are expected to possibly reflect processes other than orbital forcing. Therefore, the construction of a high-resolution age model is essential to further investigate this possibility.

2. GEOLOGICAL SETTING

The Teruel Basin (Comunidad de Aragón, Teruel Province, NE Spain; Fig. 1) is a complex of axially linked, NNE–SSW oriented half grabens developed on top of the Central Iberian Range. After the main Paleogene constructive phases of the Iberian Range, the Teruel Basin formed as a result of reactivation of Mesozoic tectonic lineaments during E–W extension that started in the early Miocene (Anadón and Moissenet 1996). The Teruel Basin is approximately 100 km long and 15 km wide, and the main phase of deposition took place between early Miocene and late Pliocene (Anadón and Moissenet 1996). Sedimentation occurred in an internally drained, semiarid, continental setting dominated by marginal to axial alluvial systems as well as lacustrine–palustrine domains of variable depth and extent (Broekman 1983; Kiefer 1988; Anadón et al. 1997). The basin fill is characterized by large-scale ($>$ 40 m) alternations of red, alluvial siliciclastics and lacustrine carbonates or evaporites (Broekman et al. 1983; Kiefer 1988; Anadón et al. 1997; Alonso Zarza and Calvo 2000). Both the Cascante and the Prado areas are located in the southern part of the Teruel Basin in the Teruel–Ademuz Subbasin (Fig. 1). This subbasin is approximately 35 km long and 15 km wide and is flanked by a Mesozoic basement, composed of Triassic fine-grained siliciclastic and evaporitic rocks and Jurassic limestones. Miocene alluvial-fan conglomerates and breccias dominate along the basin margin and grade laterally, within a few kilometers, into fine-grained distal fan to mudflat and shallow lacustrine deposits in the more central part of this relatively small subbasin (Kiefer 1988). The Prado area is accessible from the N-330 from Teruel to Ademuz, via a dirt road east of Villastar (040°12′18″ N, 001°07′22″ W).

In the Prado area ($\sim$ 5 km^2), a dominantly siliciclastic succession is situated stratigraphically between the top of the Libros Gypsum Unit exposed in the southern part of the area (Anadón et al. 1997; Ortí et al. 2003), with an approximate age of 11 Ma (Abdul Aziz et al. 2004), and a red mudstone unit with intercalated fluvial conglomerates in the northeast (Fig. 1). The sediments of the succession consist of red to yellow-orange mudstones, calcretes, and shallow lacustrine to palustrine limestones. The lateral continuity of exposures allows us to trace individual carbonate beds and mudstone color patterns over hundreds of meters from the northwest to the southeast across the area, while lateral trends from northeast to southwest can be studied over only tens of meters (Fig. 1). To the west, conglomerates and sandstones that were sourced by an adjacent alluvial fan along the western basin margin interfinger with the lower part of the stratigraphic succession in Prado. Paleocurrent directions indeed indicate a western to northwestern source. The fluvial conglomerates delimiting the top of the Prado section instead indicate a source in the north.

3. METHODS

3.1 General Methods

Stratigraphic thickness was measured by a Jacob's Staff. The lithological colors in this study comprise brownish black (7.5YR 3/1), dark red (10R 3/4-6), red (10R 4-5/6-8), (light) orange (2.5YR 6-7/6-8), pale orange (5YR 8/4), yellow orange (10YR 7-8/8), light-yellow orange (10YR 8/3-4), pale yellow (2.5Y 8/3-4), gray (5Y 4-6/1), and light gray (2.5Y 7-8/1) (Munsell 1999). The qualifier *centimeter-scale* refers to a size between 1 and 3 cm, *millimeter-scale* to between 1 and 10 mm, and *sub-millimeter-scale* to less than 1 mm. Calcrete classification follows Machette (1985), as also applied by Wright and Tucker (1991). No petrographic tools were used to further distinguish between calcrete origins. Approximately 30 thin sections of palustrine limestones were studied for microfacies analysis.

3.2 Magnetostratigraphic Methods

The Prado section was sampled at average stratigraphic intervals of 35 cm (range 10 cm to 1 m). Samples were drilled using a water-cooled, electric drill powered by a portable generator. The characteristic remanent magnetization (ChRM) was determined by thermal demagnetization, using incremental heating steps of 20 and 30°C, carried out in a laboratory-built shielded furnace. The natural remanent magnetization (NRM) of 170 samples was measured on a vertically oriented 2G Enterprises DC SQUID cryogenic magnetometer (noise level 10^{-7} A/m) in a magnetically shielded room at the Niederlippach paleomagnetic laboratory of Ludwig–Maximilians University Munich, Germany. The NRM of the other 204 samples were measured on a horizontal 2G Enterprises DC SQUID cryogenic magnetometer (noise level 3×10^{-12} A/m) at the Paleomagnetic Laboratory Fort Hoofddijk, Utrecht University, the Netherlands, following the same analytical procedure. Demagnetization results are plotted on orthogonal vector diagrams (Zijderveld 1967), and ChRM directions are calculated using principal component analysis (Kirschvink 1980).

4. SEDIMENTOLOGY

4.1 The Prado Section

A continuous sedimentary log of the Prado section (PS; Figs. 1, 2) was recorded. Due to the structural dip towards the NNE, the section is a composite of two transects; the lower transect "A" trends laterally towards the ENE, and the upper transect "B" to the north (Fig. 1). The overall section is 155 m thick, and its base (040° 15′ 17″ N 001° 08′ 45″ W) is located about 730 meters SW from its top (040° 15′ 47″ N 001° 08′ 12″ W). The log represents the complete stratigraphy in the Prado area, which has been cross-checked by several logs in lateral transects. Moreover, the

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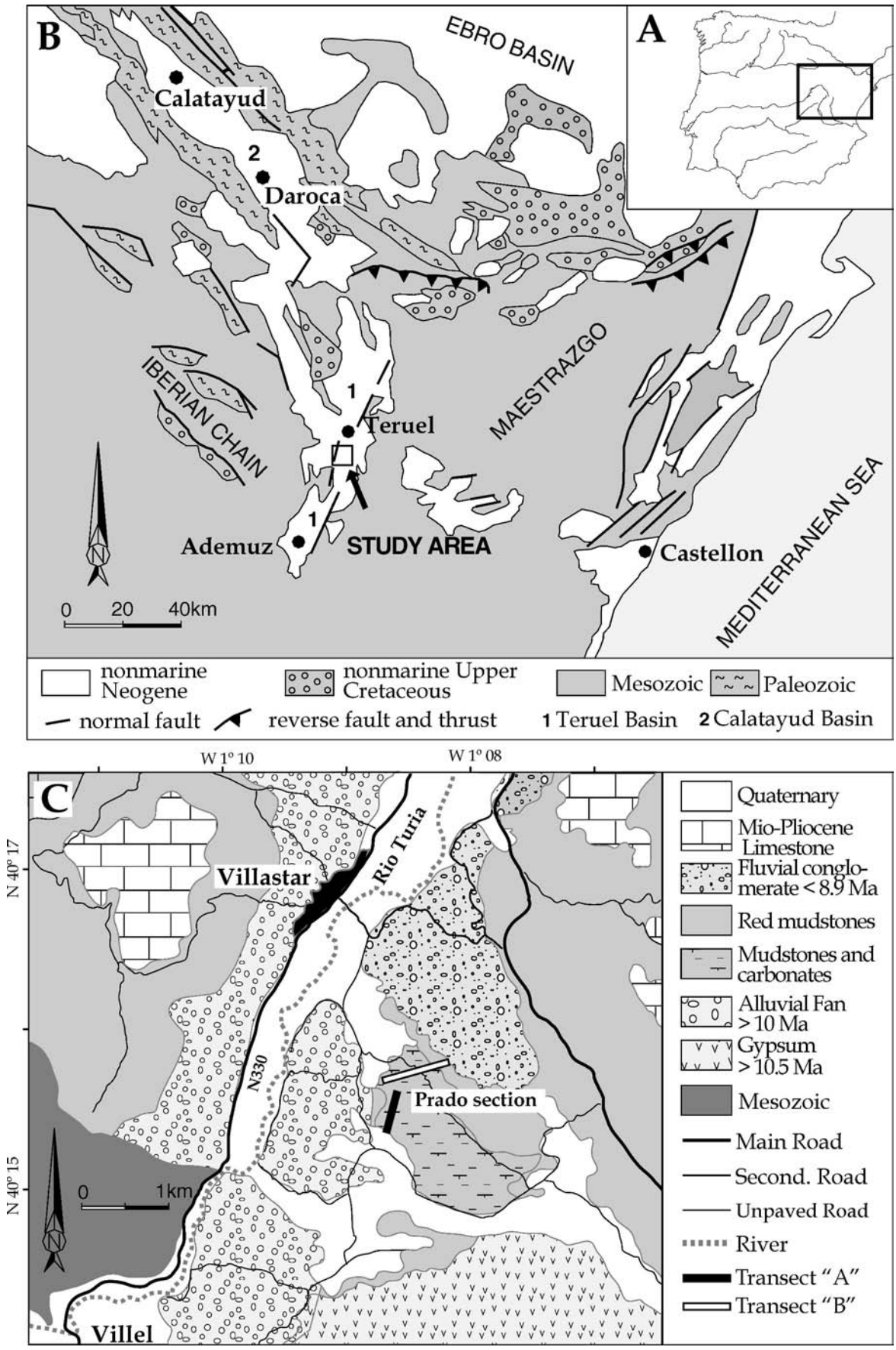


FIG. 1.—Geographic location of the Prado area within **A)** Spain and **B)** the southern part of the Teruel Basin. **C)** Schematic geological map of the Prado area. The position and extent of the two transects that make up the sedimentary log of the Prado section are indicated.

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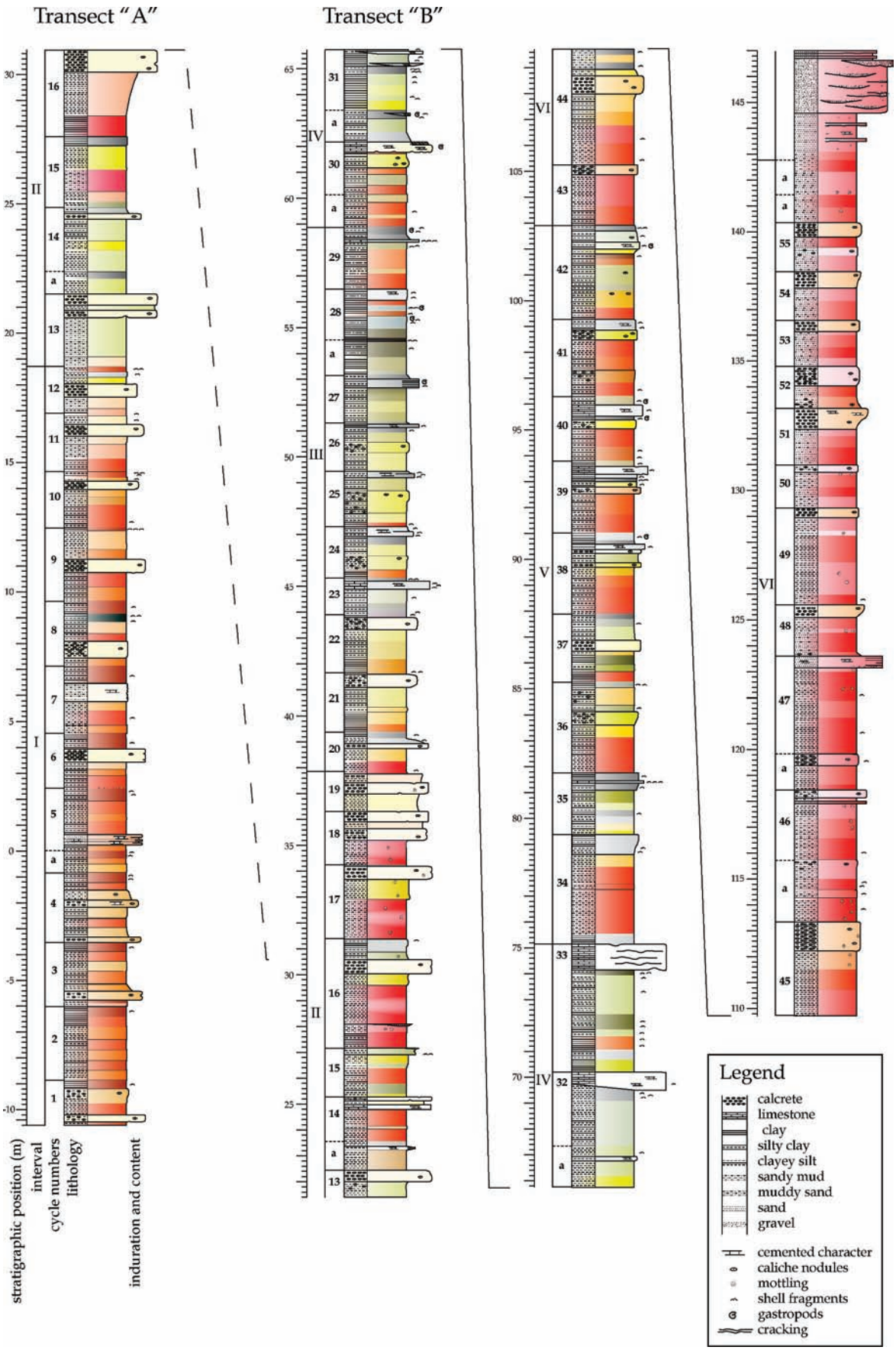


FIG. 2.— Detailed lithological log of the Prado section along the "A" and "B" transects. Stratigraphic positions, interval numbering, and cycle numbering are indicated. The dashed line indicates the correlation between the two transects. Solid lines indicate continuation of the log.

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outcrops are excellently exposed, allowing intervals to be traced laterally over hundreds of meters.

Based on distinct facies associations, the section was subdivided into six intervals, labeled I to VI (Fig. 2). In total, 55 lithofacies cycles were counted and numbered from bottom to top (Fig. 2). Ten additional cycles were counted, with less distinct lithological characteristics; these are coded with an additional "a" following the number of the cycle they are in. The sedimentological characteristics of each interval (I to VI) are reported below.

4.1.1 Interval I—Cycles 1–12.—This interval is characterized mainly by dark red to red (clayey) siltstones with meter-scale intercalations of carbonate-enriched beds, with a few orange sandy and dark-red clayey mudstone horizons.

The dark red to red clayey siltstones have a homogeneous character, commonly display millimeter to centimeter gray mottling, and occasionally show vague yellow-orange mottling. Dispersed millimeter-scale carbonate nodules occur. Gradual as well as sharp transitions occur to the meter-scale intercalations of the dark-red clayey mudstone horizons. These occur at a relatively regular basis every 2.5 to 3 meters and contain small (< 2 mm), thin-walled shell fragments. In most cases, the mudstones display sparse millimeter-scale, and occasionally centimeter-scale, gray and infrequently yellow-orange mottles and contain few, dispersed carbonate nodules.

The carbonate-enriched beds (in cycles 1, 3, 4a, 6, and 8 to 12; Fig. 2) are between 20 and 60 cm thick and are mainly pale-orange to orange with few light-gray mottles and have a dominantly silty matrix. These are relatively indurated, massive nodular beds that are occasionally horizontally brecciated at their tops. To the top of this interval, these beds are light-yellow-orange and have more frequently a very-fine sand matrix. We interpret these beds as calcretes that resemble Stage II and III calcretes according to the classification of Machette (1985). Levels slightly enriched in carbonate occur between these calcretes, with caliche nodules ranging from 5 to 20 mm in size, located just below or above the calcretes, but also as individual levels within the clayey siltstones. The carbonate-enriched beds, and to a lesser extent the calcretes, present a high lateral variability, changing in thickness over tens of meters or disappearing completely, suggesting a possible relation with grain-size distribution or indicating lateral variability of the paleoenvironment.

In the logged transect, two pale-orange, calcareous, fine-grained sandstones occur that have a clearly defined bottom and top and internally a thinner-bedded, centimeter-scale planar structure (cycles 5 and 7; Fig. 2). These sandstones are enriched in carbonate and appear as indurated levels. Here, the complete stratigraphic framework of this interval has not been worked out, because it is beyond the purpose of this study. Also, their occasional occurrence does not seem to have a consistent relation with the cyclic sedimentary dynamics observed.

Dark-red and, in cycle 8, gray mudstones with shell fragments occur regularly at an average stratigraphic distance of 2.5 m. In the lower part of interval I, these cycles are around 3 m thick, while in the upper part their thickness is less than 2.5 m. Calcretes and carbonate-enriched beds in the lower part of this interval do not reveal a consistent stratigraphic relation with these cycles. Higher up, above cycle 9, calcretes occur directly below the dark-red or gray clay layers. Lithofacies cycles in interval I were counted on the basis of the occurrence of dark-red mudstones.

4.1.2 Interval II—Cycles 13–19.—This interval includes the overlap between transects "A" and "B" that comprises cycles 13 to 16 (Figs. 3A, 2). Interval II is characterized by cycles consisting of red, orange, and light-yellow-orange, occasionally sandy, siltstones; light-gray, calcareous, clayey siltstones; limestones 20 to 80 cm thick; and gray to light-gray silty claystones.

The red sandy siltstones display sparse millimeter-scale light-gray and some orange-yellow mottling. The clayey siltstones below the limestone beds display variable thickness and show millimeter- to centimeter-scale yellow-orange mottling, which can intensify closer to the limestones.

Limestone beds occur as nodular massive, relatively smooth calcretes. These have a light-yellow-orange color and display a relatively gradual base, which occasionally undulates smoothly, and a sharp(er) top. The matrix consists of silt- to sand-size grains, as well as one thin pebble layer in the bed of cycle 13. This bed is traced as the lateral equivalent of a thick conglomerate bed. Yellow centimeter-size mottling is common, and occasionally red millimeter-size spots occur. Also, more tabular limestone beds occur with sharp bases and tops. These tabular beds display silt-size matrix, yellow mottles, and have abundant vertically oriented rhizocretions. These beds resemble palustrine limestones.

Above the limestones, fining-upward trends from siltstone to silty claystone occur, accompanied by more gray colors and an increase in organic-matter content, yellow millimeter-scale mottles, and occasionally millimeter-scale shell fragments.

This interval comprises 10 or 11 lithofacies cycles, consisting of a mudstone and a limestone or calcareous part. Cycle thickness varies from 0.9 m (cycle 13a) to 4.2 m (cycle 16), with an average of 2.5 m.

4.1.3 Interval III—Cycles 20–29.—This interval is characterized by light-gray to light-yellow-orange mudstones with sparse yellow-orange mottling and meter-scale intercalations of calcrete beds and light-gray marls and silty limestones (Fig. 3C).

The calcretes are 20 to 50 cm thick and light-yellow-orange in color, with yellow-orange mottles in a silty matrix (cycles 20 to 22 and 24 to 26; Fig. 2). Their appearance varies from a dispersed and cemented nodular to nodular massive character with more gradual bases and relatively sharp tops. They reach Stage I to III according to the calcrete classification of Machette (1985). Laterally, calcrete beds grade from loose nodule layers to cemented beds, and reverse.

Calcretes in the lower part of this interval (cycles 19 to 22; Fig. 2) are directly overlain by brownish-black calcareous mudstones that contain millimeter-scale to centimeter-scale shell fragments. These mudstones are in turn overlain by the light-yellow-orange mudstone. Higher up in this interval (cycles 23 to 26), the first 50 to 75 cm above the calcretes is characterized by light-gray marls and (light-gray) silty carbonates. Above cycle 26, calcretes are absent, and thicker (up to 40 cm), gray marls or light-gray muddy carbonates occur (cycles 26 to 29), rich in shell fragments and often complete gastropods. The muddy carbonates are soft, dusty, and poorly lithified, and display sparse millimeter-scale yellow-orange mottling, ranging in thickness between 10 and 40 cm. The thicker beds (e.g., cycles 23 and 27) also show vertically oriented, centimeter-scale root traces and, especially at their tops, horizontal cracks. Tens of meters eastwards from the logged transect, the limestone beds of cycles 23, 26, 27, 29 and 30 are much thicker, reaching up to 60 cm. These limestones occur stratigraphically associated with intervals of shell-bearing (light-)gray calcareous siltstones. Above these limestones, pale-yellow siltstones occur with red mudstones within centimeters above the carbonate interval in cycles 20, 24, 25, 28, 29.

Ten or eleven regular lithological repetitions occur within Interval III, on average every 2.0 m (range 1.6 to 2.5 m). As described above, the position of the calcretes within the lithofacies cycles changes trough the interval. Cycle boundaries are defined at the tops of the gray, shell-fragment-rich siltstones or carbonates.

4.1.4 Interval IV—Cycles 29a–33.—In interval IV, light-gray mudstones prevail, with varying amounts of yellow-orange mottling and meter-scale intercalations of gastropod-rich, light-gray, muddy limestones and (dark-)gray marls.

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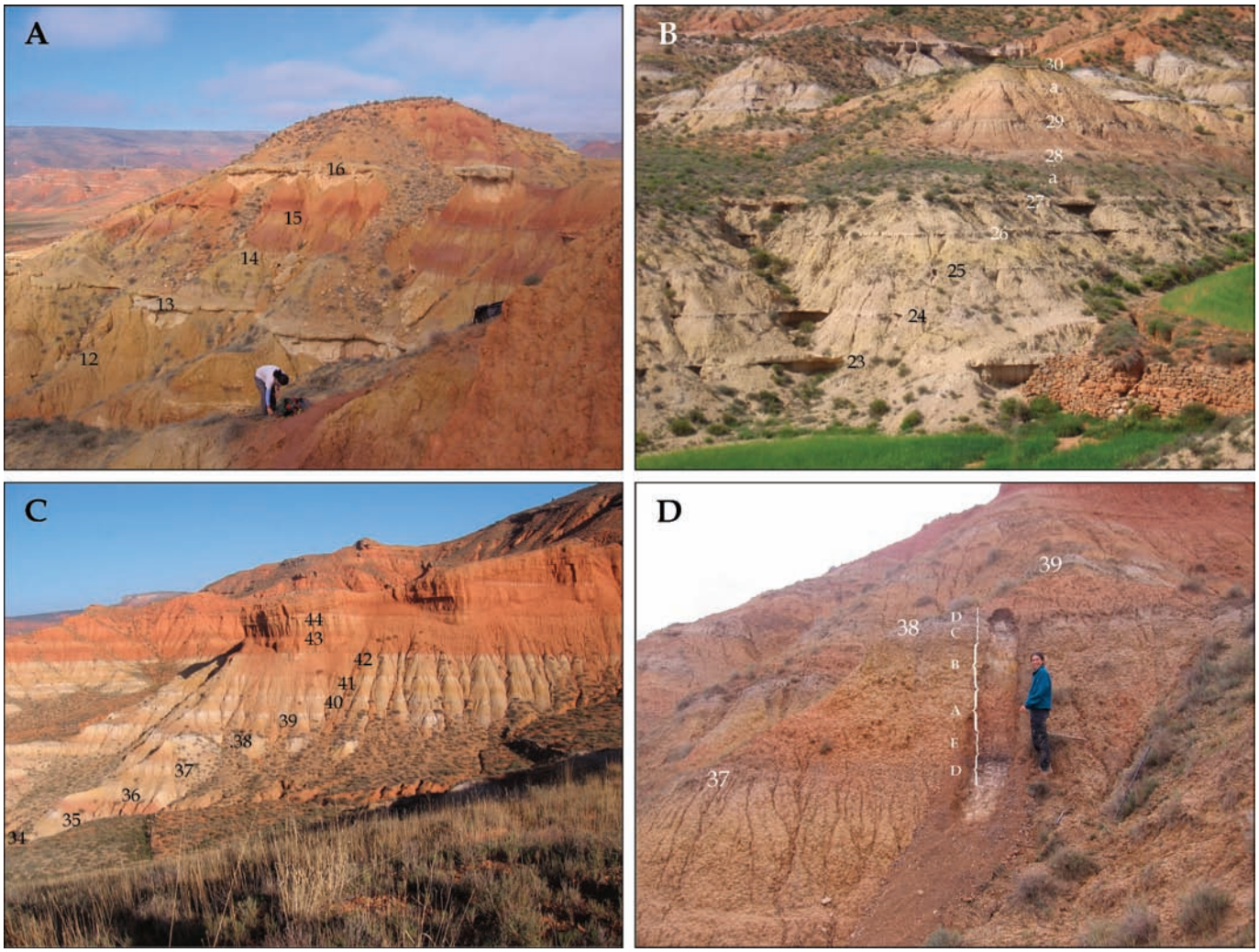


FIG. 3.—Photographs of the Prado section along the logged transects. See Figure 2 for scales. Indicated cycle numbers are positioned on the most indurated layers, all carbonate-rich lithologies. **A)** Cycles 12 to 16 in the top of the "A" transect. **B)** Cycles 23 to 30 in the "B" transect. **C)** Cycles 34 to 44 (with cycles 45 and 46 vaguely visible to the right) of the "B" transect. Above cycle 42, the second shift to well-drained floodplain sediments is visible. On top, an angular unconformity with Pleistocene conglomerates is present. **D)** Cleaned surface across cycle 38 in the "B" transect. Lithologies A to E are indicated (Fig. 4).

Light-gray mudstones display millimeter-scale yellow-orange mottling that increases in intensity and size (up to centimeter scale) just above the gray marls and limestones (cycles 28a, 28, 26a, 26).

The limestones range from 10 cm up to 1 m in thickness, and pinch out over a distance of a few meters, mainly towards the basin margin in the west and locally in other directions as well. Towards the basin center in the east, most limestone and stratigraphically associated (dark-)gray marl intervals become thick and prominent limestone beds. In the logged transect, the basal and top transitions of the beds are rather sharp. Mostly, they show gentle centimeter-scale undulations both at the bases and within the beds, and, locally, centimeter-scale bedding. Bed tops commonly feature horizontal cracks. The beds are soft and porous, with relatively much organic matter, and often full of complete and fragmented gastropod remains. Sparse millimeter-scale yellow-orange mottling and locally centimeter-scale root traces are present. The limestone beds are stratigraphically associated with (dark-) gray marls that are rich in very fine-grained organic matter, most probably plant debris, and fragmented and complete gastropod shells. The organic-rich levels in interval IV are occasionally rich in small-mammal remains, and sparse large-mammal remains have been found.

In this interval, cycle boundaries are defined at the top of the gray mudstone or limestone beds. Cycle thickness varies between 3.3 to 5.0 m for the 4 well-defined cycles (cycles 30, 31, 32, 33). In between these cycles, three less clear cycles are present (cycles 29a, 30a, and 31a; Fig. 2).

4.1.5 Interval V—Cycles 34–42.—Interval V is dominantly characterized by red mudstones with intercalations of between 75 and 150 cm of light-yellow-orange calcretes, (dark-)gray calcareous mudstones, and light-gray limestones of meter scale (Figs. 3B, D).

The red mudstone displays sparse millimeter-scale yellow-orange and light-gray mottling and occasionally millimeter-scale dispersed caliche nodules. The basal part of this mudstone is often dark red changing upwards to red, while more intense yellow-orange mottling usually increases towards the top, resulting in orange colors.

The intercalations generally start with a calcrete bed, 20 to 50 cm thick, that varies between loose nodule levels to more cemented nodule levels, and occasionally massive nodular beds. The calcretes are light yellow to yellow, have a silty to muddy matrix, occasionally display vertical arrangement of nodules resembling root morphologies, and show relatively gradual bases and sharp tops and upward-increasing carbonate

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content. Nodules vary in size from sub-millimeter to centimeter scale and are subspherical. The calcretes reach Stage II to III (cycles 36 to 41; Fig. 2) according to the calcrete classification of Machette (1985).

The calcretes are followed by a (dark-)gray calcareous mudstone with sparse shell fragments. In some cycles, a (dark-)gray marl rich in shell fragments occurs, with local intercalations of a light-gray limestone (cycles 34, 35, 38 to 40, and 42). The thickness of the gray calcareous mudstone ranges between 20 and 125 cm.

The intercalated limestone beds range in thickness between 10 and 40 cm, are light gray, have undulating bases, and contain shell fragments and infrequently complete gastropod shells. The limestone beds in this interval have sedimentological characteristics similar to the limestones of cycles 30 to 33 in interval IV, though the latter are thicker. Locally, the limestone pinches out or increases in thickness laterally over a few meters towards the east as well as west.

Interval V comprises nine sedimentary cycles with an average thickness of 3.3 m, and ranging between 2.4 and 4.3 m in thickness. The repetitive pattern of sediments is very regular in this interval, and cycle boundaries are easily defined at the bases of the red mudstones.

4.1.6 Interval VI—Cycles 43–55.—A distinct change occurs above interval V (cycle 42; Figs. 2, 3B), from dominant (light-yellow-)orange and red mudstones with intercalated gray mudstones and limestones, to dominant orange silty and sandy mudstones (Fig. 3B). This transition was traced laterally from east to west over a distance of 0.5 km, which is the limit of the outcrop exposures.

The interval above this shift, interval VI, is characterized by dominance of orange, fine-sandy siltstone and light-orange, silty, fine sandstone. Sparse, millimeter-scale light-gray mottling occurs, with occasional levels of centimeter-scale light-gray mottling and sparse millimeter- to centimeter-scale carbonate nodules. On meter scale, intercalations of calcrete beds 40 to 120 cm thick and fine to pebbly sandstones occur.

The calcrete beds have a silty to fine sandy matrix and millimeter-scale (light-)yellow-orange mottling, and are usually poorly cemented loose nodule levels and occasionally better-cemented nodular beds. In poorly cemented nodule layers, a dark red matrix with millimeter-scale yellow mottling occurs, especially near the tops of the beds. In redder intervals, millimeter- to centimeter-scale light-gray mottling occurs. The bases of the beds are gradual, and their tops are relatively sharp. According to the calcrete classification, these calcretes represent Stage 1 to 2 (Machette 1985).

In cycle 44, the calcrete is covered by a gray, organic-rich, fine-sandy mudstone, while in other cycles (cycles 43, 45, 46, and 48) calcretes are followed by red sandy siltstones occasionally displaying sparse millimeter-scale shell fragments.

The fine-sandy to pebbly intercalations are cemented by carbonate and fine slightly upward. At the base of the sandstone bed in cycle 47, a 50-cm-thick conglomeratic lobe pinches out within 10 meters both to the east and to the west. This conglomerate is characterized by a clast-supported fabric of subrounded pebbles and rare cobbles, a poorly sorted muddy to coarse sandy matrix, and a relatively high matrix-to-clasts ratio. The associated sandstone beds consists of poorly sorted, fine to very coarse sand with plane-parallel lamination and sparse, floating pebbles.

Above interval VI, a coarsening-upward towards a conglomerate 2 to 5 m thick is present. Locally, the base of the conglomerate incises up to 3 meters in the sandy interval below. The conglomerate consists of several ~ 30-cm-thick fining-upward beds. The pebble clasts vary in size from 1 to 30 cm, with an average of 4 cm. The subrounded clasts are densely packed in a fine to very coarse sand matrix.

Cycles in interval VI have an average thickness of 2.9 m but show a wide range from 1.6 to 5.2 m. Depending on two less well defined cycles (45a and 46a), 13 to 15 cycles are present excluding the two potential cycles above cycle 55. The lower part of this interval consists of relatively thick cycles thicker than 3 m (cycles 44 to 47, and 49), while towards the top the average cycle thickness is only 1.8 m (cycles 50, and 52 to 55).

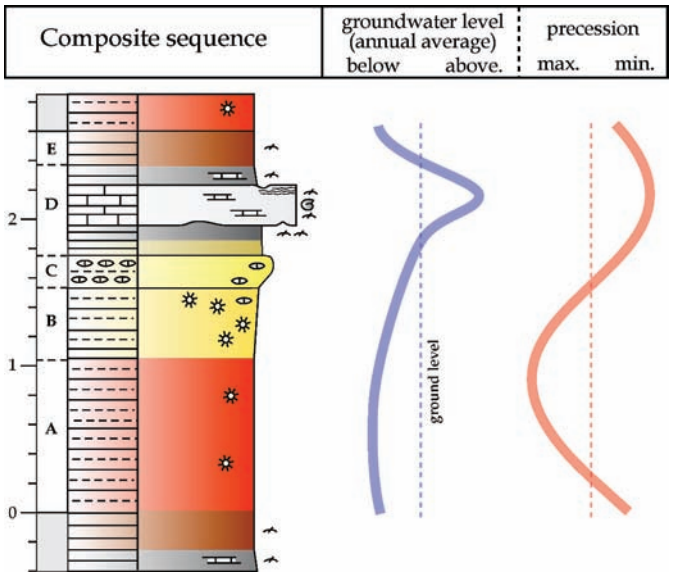


FIG. 4.— Lithological characteristics of the composite lithofacies cycle in the Prado section (see Fig. 2 for legend and the section 4.2 on description and interpretation of facies A to E). Scale is in meters. To the right, interpretation in terms of local, relative annual averaged groundwater level and the precession cycle are given. Rd for red, gy for gray, or for orange, d for dark, and l for light. This figure is in color in the digital version of the article.

4.2 Composite Sedimentary Cycle

The persistent, meter-scale lithofacies cycles in the Prado section generally consist of a red and/or (light-)yellow-orange mudstone part, mainly encompassing the lower, thicker fraction of the cycle, and a carbonate-rich part (Figs. 3D, 4) represented by light-yellow-orange calcretes and gray organic-matter-rich muddy limestones, or marls. Cycle boundaries were consistently placed at the transition from dominantly calcareous to dominantly siliciclastic intervals (see above for details). In total, 55 cycles show a well-defined lithofacies transition at their tops (solid lines in Fig. 2), while 10 additional cycles show more diffuse transitions or less distinct lithologies (dotted lines in Fig. 2). The average thickness of a sedimentary cycle in the Prado section is 2.6 meters, with a standard deviation of 0.9 meters and extremes ranging from 1.2 to 5.2 meters. Three intervals with thick cycles ($>$ 3.5 m) occur in the section: cycles 16 and 17 in interval II, cycles 30 to 33 in interval IV, and cycles 46, 47, and 49 in interval VI (Fig. 2). In order to facilitate interpretation and discussion of facies and the variable sedimentological expressions of stratigraphic intervals, an idealized composite lithofacies cycle was constructed (facies A to E; Figs. 3D, 4). This cycle consists of five distinct facies, superimposed according to their recurrent vertical transitions in the different types of cycles discussed above. This composite cycle is thus not found as a real stratigraphic unit at outcrop, but it reliably represents the overall facies variability, and especially facies transitions, which are recognized consistently throughout the section.

4.2.1 Facies A.—Description.—The composite cycle starts with a red, occasionally orange or dark-red, massive and homogeneous silty mudstone (Fig. 4), displaying different amounts of sparse light-gray, millimeter- to centimeter-scale mottling. Rare millimeter-scale carbonate nodules are found. Infrequent millimeter-scale shell fragments occur, especially at the bases of the mudstone beds. The mudstones lack primary

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depositional structures, well-developed pedologic (soil) horizons and structures, and internal discontinuities and erosional features at their bases and tops. Lithological transitions at their bases are usually relatively sharp, whereas transitions to overlying lithologies are more gradual.

Interpretation.—The massive and homogeneous red mudstones display the characteristics of a well drained mudflat deposit with very weak pedogenic overprint (Kraus and Hasiotis 2006). The sparse, light-gray mottling indicates weakly reducing conditions due to short periods of seasonal wetting (Kraus 2002; Kraus and Hasiotis 2006) or localized biological activity. The scarce rhizocretions and the absence of bioturbation point to poor colonization by vegetation or other organisms (Huerta and Armenteros 2005). The rare carbonate nodules, interpreted as incipient caliche, indicate precipitation of calcium carbonate in the Bk horizon of a calcic soil profile (Retallack 2001). Lack of well-developed soil profiles and structures indicate only minor pedogenesis within the deposit. A semiarid climate could have been a significant cause of poor pedogenesis, although the overall homogeneity of facies vertically through the beds also suggests that time was a significant factor. The slow development of a mature soil profile was probably hindered by continuous, relatively high aggradation rates (Buurman 1980). Incipient pedoturbation, desiccation, and compaction possibly removed any original structures (cf. Wright and Marriott 2007). The fine-grained sediments were probably deposited by distal sheetwash during waning-flow stages of major flood events from the adjacent basin margin where these lost capacity over the flat, sparsely vegetated distal surfaces (Abels et al. 2009).

4.2.2 Facies B.—Description.—The transition from facies A to B is usually gradual. Facies B consists of orange to light-yellow-orange, massive, silty to sandy mudstones. These mudstones display abundant centimeter-scale yellow-orange mottling, minor millimeter-scale light-gray mottling, and a generally higher content in dispersed carbonate nodules and rhizocretions than in facies A. In some cycles, the sediment is rather uniform in color, whereas in others it has a multi-colored, patchy appearance with red, yellow-orange, light-yellow-orange, and light-gray mottling. The abundance of millimeter- to centimeter-scale carbonate nodules and intense yellow-orange mottling tends to increase towards the top of facies B. Similarly to facies A, facies B is also characterized by internal homogeneity and the absence of primary sedimentary structures at outcrop scale. The boundary between facies A and B is defined arbitrarily within the composite cycle, whereas in the field it is usually diffuse over an interval with characteristics intermediate between A and B.

Interpretation.—The sedimentary processes by which facies B was formed are interpreted as fairly similar to those for A. The most significant difference is the slightly coarser grain size and slightly higher carbonate and rhizocretion content in facies B, accompanied by a greater size and color intensity of mottled domains, and the general occurrence of (light-)yellow-orange mottling. The upward increase of yellow-orange mottles from the red mudstones of facies A to the orange mudstones of facies B is interpreted as an increase of (seasonal) wetting of the soil profile (Kraus 2002; Kraus and Hasiotis 2006; PiPujol and Buurman 1997). The slightly coarser grain size and increased carbonate content and pedogenesis might be indicative of a more frequent or persistent availability of water. The prevalence of Jurassic and Cretaceous carbonate formations in the Mesozoic catchment areas led to a high availability of dissolved carbonates in both surface runoff and groundwater. The increase in overall carbonate content from facies A to facies B might thus be related to rising availability of water via groundwater and/or surface runoff. Towards the top of facies B, these (seasonally regulated) soil-forming processes were further enhanced. The presence of yellow-brown colors in red soils and the dispersed presence of carbonate nodules suggest moderately well-drained conditions with a short period of seasonal wetting (Kraus and Hasiotis 2006).

4.2.3 Facies C.—Description.—Above facies B, calcrete beds 20 to 60 cm thick occur that display gradual bases and relatively sharp tops. The beds are commonly light-yellow-orange with yellow-orange and occasionally light-gray mottles. Beds vary from noncemented carbonate nodule levels, to better cemented nodular levels, and finally to indurated massive nodular beds with a smooth appearance.

Weakly-cemented nodule levels display (dark-)red matrix and millimeter-scale to centimeter-scale light-gray mottling with sparse yellow-orange mottling. Better-cemented beds dominantly display centimeter-scale yellow-orange mottling with infrequent millimeter-scale light-gray mottles. The matrix is usually silty to, infrequently, fine sandy. Carbonate nodules are millimeter- to centimeter-scale and subspherical, and may display a vertical arrangement resembling root morphologies. At the tops of the beds, horizontal structures may be present, either as millimeter-scale cracks in cemented beds or as horizontally oriented nodules in less-cemented beds. Laterally, less-cemented beds may grade into more indurated beds and vice versa. Also, the stratigraphic thickness of individual beds may vary laterally, including smooth undulation of the bases of the beds. According to the calcrete classification of Machette (1985) as followed by Wright and Tucker (1991), the calcrete characteristics would fit Stage I or II for the incipient weakly cemented nodule levels to between Stage III and IV for well-cemented massive nodular beds. No horizontal platy, tabular structures characteristic of a full Stage IV calcrete are observed.

Interpretation.—Carbonate precipitation in the Stages I to IV calcretes likely occurred due to calcium-carbonate saturation in the capillary fringe zone above the groundwater table or in the Bk horizon of a calcic soil profile or due to a mix of these pedogenic and phreatic processes (Alonso Zarza 2003; Huerta and Armenteros 2005; Retallack 2001; Wright and Tucker 1991). Climate was semiarid and seasonally dry with an annual precipitation estimated between 500 to 550 mm (Van Dam 2006), supporting the occurrence of pedogenic calcretes; however, other climate(s) are reported to support calcrete formation as well. The original red mudstone matrix and the occasional lateral gradual changes from weakly cemented to more indurated beds suggest that the (facies C) calcretes did not (principally) originate from calcretized lacustrine or palustrine carbonates (Huerta and Armenteros 2005; Wright and Tucker 1991). Nevertheless, the carbonate-rich basin margins and underlying lithology suggests that dissolved carbonate was abundantly present in the Prado paleoenvironment. Also, the relatively impermeable sub-surface, due to the presence of multiple well-cemented calcretes in the Prado succession and the gypsum formation below, may have resulted in a relatively poor drainage, giving evapo(transpi)ration a significant role, thereby redistributing the abundant dissolved carbonate and facilitating relatively rapid calcrete development (Wright and Tucker 1991). The availability of water instead of time or (dissolved) carbonate might have been the critical factor in controlling carbonate deposition in calcrete beds in the Prado paleoenvironment. We interpret the gradual change from facies A to B and finally to C as a gradual increase in water saturation of the soil profile, in accordance with a lateral facies model of Huerta and Armenteros (2005) for Miocene sediments in the Duero basin. In their model, lateral coexistence of reddish-brown mudstones, scattered carbonate nodules, nodular calcrete facies, and massive calcrete is shown in a transect from well-drained distal alluvial floodplain, to less well-drained distal floodplain, and finally to more inundated areas near the lake margin.

4.2.4 Facies D.—Description.—The lower portion of facies D, over a variable thickness between 0 and 60 cm, represents characteristics transitional from facies C. This transitional interval constitutes gray

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calcareous mudstones with light-orange-yellow mottling. This interval is covered by dominantly light- to dark-gray marls and muddy limestones, locally intercalated with 20-cm-thick light-gray limestone beds that have a maximum thickness of 1 m. Mudstones and marls contain abundant millimeter- to centimeter-scale shell fragments and complete gastropod remains. Thin (< 10 cm) organic-rich layers are present, usually containing coarse plant debris and remains of small mammals, and occasionally large mammals.

The intercalated light-gray limestone beds have sharp undulating bases, are weakly indurated, and contain a relatively high amount of fine siliciclastic components. In many cycles, the limestones are rich in dispersed organic matter and intact or fragmented gastropod remains. Besides the undulating bases, slightly undulating internal surfaces are locally evident within the limestone beds, especially in the thicker ones. Some beds pinch out laterally over distances of a few meters, but usually they extend over 50 meters at outcrop, and some are traceable over hundreds of meters towards the more central parts of the basin, farther east from Prado. The tops of the limestone beds usually feature horizontal cracks, while sparse rhizoturbation is present throughout the beds, and more frequently within their upper parts.

Interpretation.—Deposition of gray, gastropod-rich, calcareous mudstones and marls indicates rising groundwater levels and occasional ponding of the surface. These conditions rapidly halted calcrete development, and caused fine clastic sedimentation to alternate with (biochemical) carbonate precipitation. Temporary reduced oxygenation is indicated by preservation of abundant organic matter, including gastropod and mammal remains, and by light-gray colors indicating the remobilization of iron (Kraus and Hasiotis 2006). Intercalated, undulating, light-gray limestones are interpreted as the result of carbonate deposition in very shallow and relatively wide, elongated depressions through which sluggish surface waters were slowly directed towards the basin center, or to local small ponds on the mudflat. Thicker limestone beds, laterally continuous over long (> 500 m) distances, are interpreted as pond and marginal lacustrine deposits. The absence of extensive palustrine features within and on top of limestones beds suggests that their deposition rates were relatively fast, leaving no significant time for reworking at the sediment–water interface in shallow-water conditions. Palustrine features are reported to develop rapidly in freshwater limestones (Alonso Zarza 2003), while desiccation is particularly expected in very shallow-water environments in a semiarid, seasonal climate (Van Dam 2006).

4.2.5 Facies E.—Description.—The topmost facies within the composite lithofacies cycle is represented by a dark-red to brownish-black, massive silty clay containing dispersed millimeter-scale shell fragments. Millimeter-scale, yellow-orange and red mottling is present. No primary sedimentary structures are evident at outcrop.

Interpretation.—The dark mudstones are interpreted as mudflat deposits, analogous to those of facies A and B. Environmental conditions did not allow more than weak pedogenesis keeping pace with aggradation of the surface. Mild pedoturbation disrupted any primary sedimentary structures but did not reach a full stage of soil formation. The vague reddish and occasionally yellow-orange mottles characterizing facies E probably indicate a return to more oxygenated conditions of the floodplain, related to a gradual shift from a poorly drained (facies D) to a well-drained (facies A) mudflat environment (Kraus and Hasiotis 2006). Within the interpretive framework of our representative sedimentary cycle, facies E is regarded as a transitional facies between D and A.

4.3 Depositional Model

The representative facies succession A to E in the meter-scale lithofacies cycles in Prado are interpreted as deepening-upwards sedimentary sequences due to variations in relative groundwater level at the site of deposition. Groundwater level and/or water availability was low on a well-drained mudflat during deposition of facies A. The subsequent rise of the water table resulted in more effective pedogenic processes, produced facies B, and eventually led to the development of calcic soil profiles or phreatic calcretes (facies C). The further rise of groundwater levels resulted in more persistent saturation up to the ground surface, with temporary development of the poorly oxygenated wet mudflat environment of facies D, in which sluggish elongated, low-energy channel bodies and small ponds developed. Occasionally, this rise led to the development of marginal lacustrine environments, especially towards the center of the basin. Finally, a lowering of groundwater level resulted in a stratigraphically rapid return to the well-drained, dry mudflat environment represented by facies A, with facies E as a transitional stage. In the composite sedimentary cycle, the facies A and B, related to groundwater-level lowstands, are referred to as the "dry" part of the cycle. Conversely, facies C, D, and E, which are related to groundwater-table highstands, are referred to as the "wet" part of the cycle. Note that with the "wet" and with the "dry" part of the cycle we refer only to the relative groundwater table in the sediment at the site of deposition and we do not insinuate any mechanism behind these changes.

4.4 Large-Scale Changes

Throughout the Prado section, intervals I to VI all show different facies characteristics that are variations of the composite cycle. Four characteristic intervals (part of intervals I, III, V, and VI) are compared (Fig. 5), in order to illustrate these major sedimentary signatures and paleoenvironmental changes. The sediments of the other two intervals, II and IV, are not elaborated here because they represent transitional intervals with thicker-than-average cycle thicknesses including less well-defined cycles. Gray lines in Figure 5 indicate cycle boundaries and thus the transition from the "wet" to the "dry" parts of the cycles. In the basal part of interval I, the stratigraphy is composed mainly of facies A, B, E, and occasionally C. In interval III, cycles are composed dominantly of facies C, D, and only rarely E, while facies A and B are absent. Higher up, in interval V, facies A and B reappear and the cycles closely resemble the composite sedimentary cycle model, displaying the whole facies spectrum from A to E. With a sharp transition above cycle 42, cycles in interval VI are again dominated only by facies A and B, with thin intercalations of facies C. The main differences between intervals I and VI are the absence of facies E and the higher lateral continuity of carbonate-rich beds in VI. Interpretations in terms of local, relative hydrological balance suggest a progressively rising groundwater table from interval I to III, followed by a subsequent lowering in two main steps: at the transition from interval IV to V (above cycle 33), and at the transition from interval V to VI (above cycle 42).

Above interval VI, the Prado section is topped by tens of meters of red, weakly pedogenically modified mud deposits with minor sandstone sheets and conglomeratic channel fills in single and multiple stories. On a decimeter to meter scale, the mudstones present rhythmic facies transitions similar to facies A and B. The lateral continuity of these deposits is occasionally interrupted by the erosional bases of coarse-clastic channel bodies and associated sandstone splays. This overlying succession is interpreted as being deposited in a poorly confined, ephemeral fluvial system sourced from the north, which is along the strike of the basin. The greatest volume of sediments is represented by overbank fines grading laterally to shallow, isolated channels that are locally filled by much coarser and poorly organized sediment, in correspondence of major floods. The presence of these erosional channels suggests higher gradient within the basin that is related to increased accumulation rates. The end of the endorheic configuration of the basin is discarded here, in as much as clearly endorheic lacustrine sediments are present up to the Pliocene in the basin.

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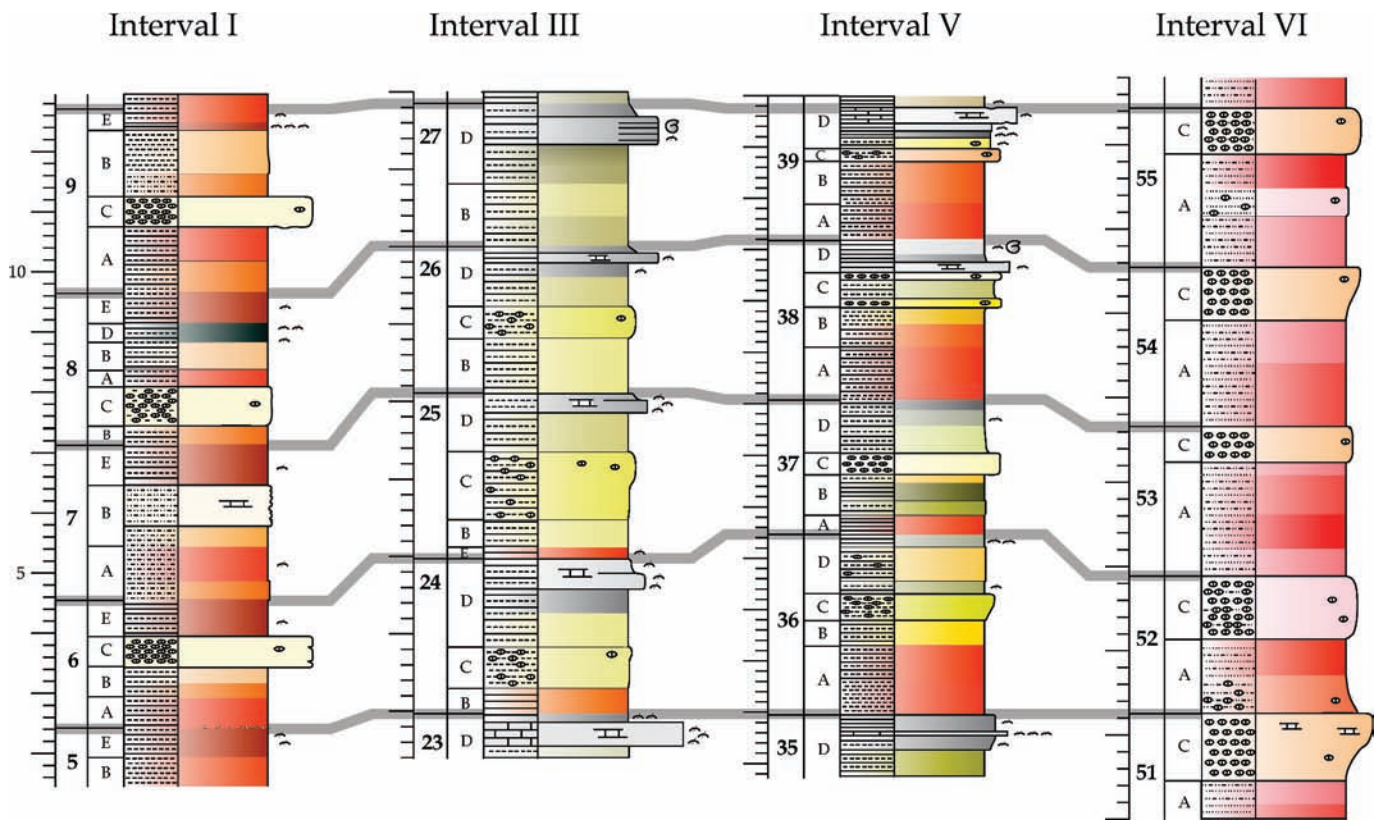


FIG. 5.—Detailed comparison of the four distinct stratigraphic intervals in the Prado section, which reveal regular meter-scale lithofacies cycles. The well-drained red mudstone facies gradually disappears from interval I to III but reappears in two steps from III to V and from V to VI (see sections 4.4 and 6.5).

5. MAGNETOSTRATIGRAPHY

A detailed magnetostratigraphic age model was established to test whether the regular occurrence of meter-scale lithofacies cycles is related to orbital climate forcing. First-order age control is derived from lithostratigraphic correlation to the well-dated time-equivalent Cascante section (Abels et al. 2009), where the magnetostratigraphy was tied to the geomagnetic time scales using biostratigraphy of small-mammal faunas (Abdul Aziz et al. 2004). The presence of the top of the Libros Gypsum Unit (Anadón et al. 1997; Ortí et al. 2003) below both the Cascante and the Prado sections indicates that they are roughly time-equivalent, which is confirmed by preliminary mammal biostratigraphic results (J. Van Dam, personal communications, 2008).

5.1 Results

The results of the thermal demagnetization for the Prado samples are of good quality (Fig. 6A–D). Initial NRM intensities vary according to facies (Fig. 7), with highest NRM intensities (7.8 to 95.4 mA/m) in the red mudstones and siltstones of facies A and E that typically dominate in intervals I, II, and VI. The lowest intensities (0.04 to 7.8 mA/m) are found in facies B and D, which contain more carbonate and dominate intervals III and IV. The calcretes of facies C show a wider range of NRM intensities varying between 0.07 and 87.3 mA/m, with highest values corresponding to the lithologies from interval VI (cycles 43–55) (Fig. 7). The Zijderveld diagrams and thermal decay curves show that the total remanent magnetic signal consists of three components. The first is a randomly oriented component that is removed between temperatures of 100°C and 120°C, and represents a laboratory-induced magnetization related to storage. The second component is removed between 210°C and 270°C; however, in a few samples this component is removed up to temperatures of 390°C (especially in cycles 49 to 51; Fig. 6C). This component has a normal polarity and is interpreted to represent a viscous overprint by the present-day Earth's magnetic field. The third component displays dual polarities and is interpreted as the characteristic remanent magnetization (ChRM) of the sediment. Three unblocking temperatures of the ChRM can be distinguished in the Prado section. The ChRM of ~ 60% of the samples is fully demagnetized at temperatures from 600°C up to 700°C, indicating unblocking temperatures for (fine-grained) hematite (Fig. 6A, C, E). Samples with these unblocking temperatures are mostly from lithologies A and B (and E), i.e., from the well-drained-mudflat deposits. The ChRM of ~ 35% of the samples is removed between 520°C and 600°C, which is a typical unblocking temperature range for magnetite (Fig. 6B, D). Samples with this temperature range include all facies A to E. The remaining 5% of the samples have unblocking temperatures between 390°C and 480°C, suggesting iron sulfides as the main carrier of the ChRM (Fig. 6F). Most of these samples belong to carbonate-rich mudstones of facies D. Susceptibility measurements indicate stable values for the samples with high unblocking temperatures (570°C and higher) with no significant change in mineral composition during the thermal demagnetization. The samples from the calcareous gray mudstones of facies D show a distinct increase of susceptibility values from temperatures around 360°C and higher, probably due to the oxidation of the iron sulfides.

The ChRM directions were calculated for more than five temperature steps in the range 270 to 700°C. The quality of the measurements and the line fitting were evaluated by visual inspection of Zijderveld diagrams and by calculating the maximum angular deviation (MAD). The magnetic polarity record of the Prado section was interpreted using MAD values smaller than 20° as cutoff. Values larger than 20° and non-interpretable samples are indicated in Figure 7 by open circles and crosses, respectively.

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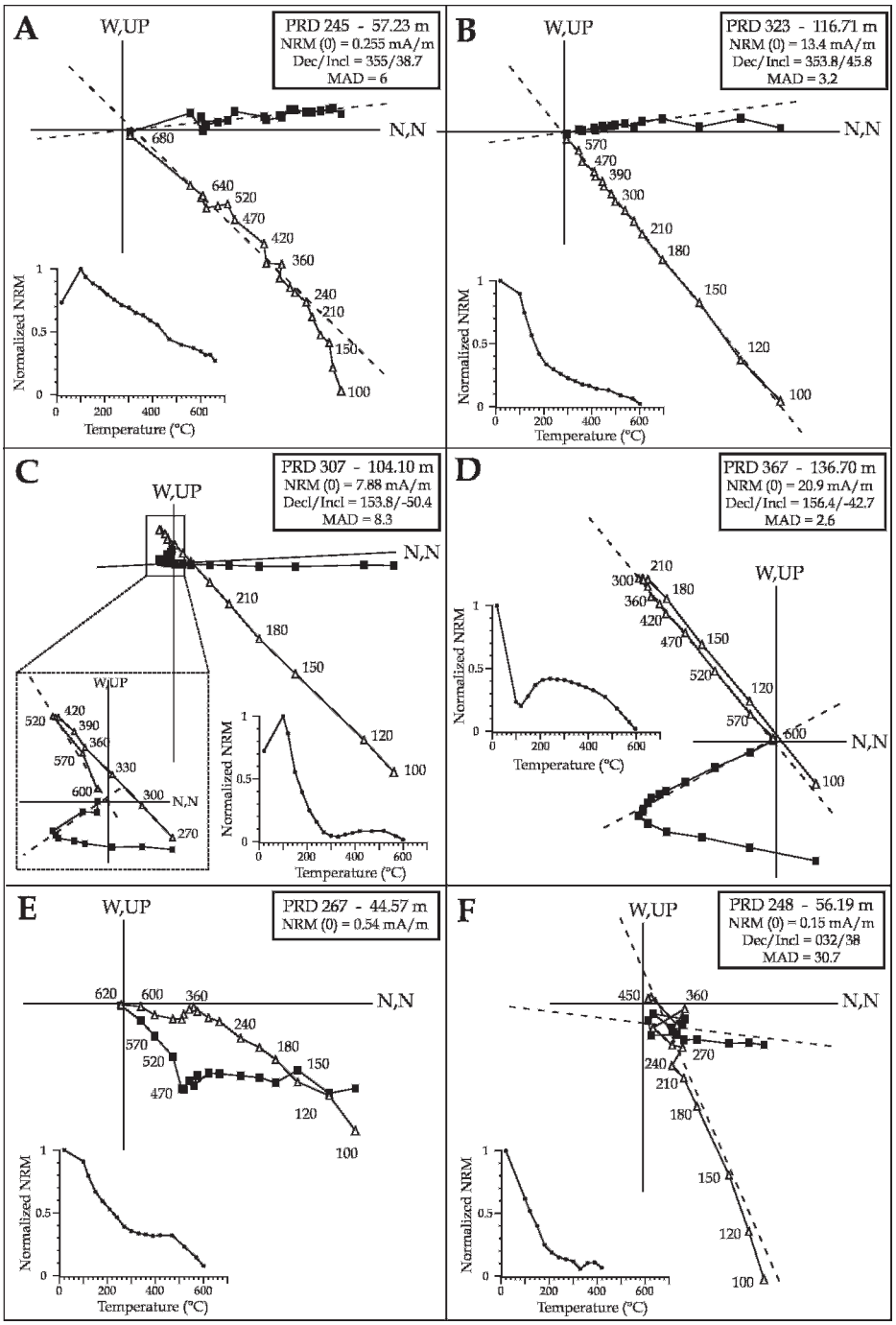


FIG. 6.—Thermal demagnetization diagrams and decay curves (inset) of selected samples from the Prado section. Open (closed) symbols denote projection on the vertical (horizontal) scale. Numbers along demagnetization trajectories indicate temperature steps in °C. Dashed lines are the interpreted declination and inclination direction of the characteristic remanent demagnetization (ChRM). MAD denotes the maximum average deviation of the interpreted ChRM direction. See text for details on A–E. Part F denotes an example with unresolved polarity.

5.2 Correlation to the ATNTS2004

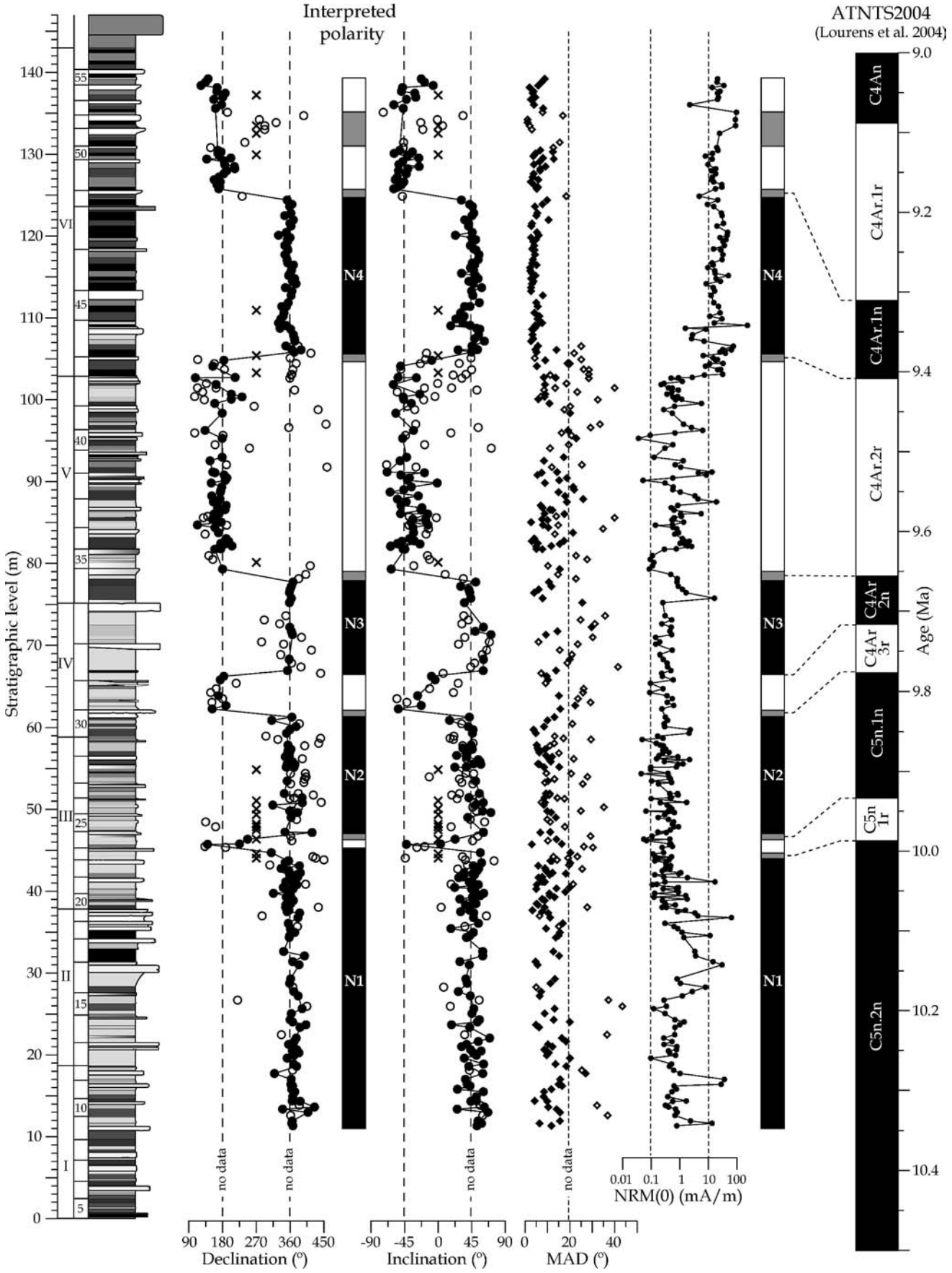
Declination and inclination data are plotted in stratigraphic order and reveal seven polarity reversals, with gray colors indicating the uncertainty intervals of the reversal level (Fig. 7). The established lithostratigraphic and time-equivalent relationships between the Prado and Cascante sections allow a straightforward correlation of the Prado polarity record to the ATNTS04 (Lourens et al. 2004). Starting from the base of the Prado section, the long normal-polarity interval N1 is correlated to chron C5n.2n. The subsequent three normal-polarity intervals N2, N3, and N4 are correlated to chrons C5n.1n, C4Ar.2n, and C4Ar.1n, respectively. The correlation of the Prado magnetostratigraphy to the ATNTS04 of Lourens et al. (2004) reveals a complete polarity record ranging from chron C5n.2n to C4Ar.1r and covering a time span of more than 1 million years, i.e., from ~ 10.3 to 9.2 Ma.

6. CYCLOSTRATIGRAPHY

6.1 Orbital Forcing

Correlation of the magnetostratigraphy with the ATNTS2004 time scale (Lourens et al. 2004) provides a detailed age model for the Prado section. In the time interval considered, the astronomical ages for polarity reversals in the ATNTS2004 were derived from astronomically tuned

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sapropel patterns in the deep marine Monte dei Corvi section in northern Italy (Hüsing et al. 2007). In Figure 8, the magnetostratigraphy of the Prado section is compared to the recently improved astronomically dated ages of polarity reversals in this deep marine section. Gray bars represent ranges of uncertainty in depth at Prado and time at Monte dei Corvi.

In the magnetic-polarity interval between the reversals at the top of chron C5n.1r and the top of C4Ar.1n, there are 24 well-defined lithofacies cycles in the Prado section with an uncertainty of half a cycle in both directions. Additionally, there are five less well-defined "a" cycles in this interval, making a total of 29 ± 0.5 cycles (Fig. 8). In the Monte dei Corvi section, the time between the two reversals is 626 ± 9 kyr (Hüsing et al. 2007). This results in a period between 25.1 and 27.0 kyr for a well-defined lithofacies cycle, and between 20.9 and 22.3 kyr when the "a" cycles are included. The latter period is suggestively close to the duration of a *climatic* precession cycle with main periods of 18.9, 22.2, and 23.5 kyr (Laskar et al. 2004). The suggestion by the age control that the meter-scale lithofacies cyclicity may be driven by precession is further elucidated.

6.2 Precession Phase Relation

In the depositional model presented in Figure 4, the paleoenvironmental interpretation of the sedimentary cycles was formulated in terms of local, relative hydrological balance. Inasmuch as precession forcing is suggested by the age model, meter-scale sedimentary rhythms controlled by groundwater levels then must be related to precession. The phase relation between changes in the regional water table and precession remains, however, unknown. The phase relation relates to two elements: the relationship between groundwater level (and, consequently, sedimentology) and local climate, and the link between local climate and precession.

The relation between groundwater table (and sedimentological processes) and local climate can be straightforward. Lower groundwater tables inferred from the development of well-drained floodplains in facies A and B ("dry" part of the cycle) might indeed relate to decreased precipitation rates, and vice versa. However, increased sedimentation rates caused by increased precipitation could have a similar effect. According to the latter hypothesis, waterlogged environments could then occur during times of sediment starvation due to decreased precipitation. These processes might have a different effect on different parts of the basin (Picard and High 1981). Increased seasonality with a short wet season is suggested by yellow-orange mottling in facies B with respect to facies A (see interpretation in the section 4.2.2; Kraus 2002; Kraus and Hasiotis 2006).

Abels et al. (2009) elaborated on the link between local climate and precession, when trying to solve the same problem for the sedimentary cycles in the nearby, time-equivalent Cascante section in the Teruel Basin. Geological data as well as climate modeling of precession extremes indicated that winter precipitation increased significantly during precession minima (and related boreal summer insolation maxima), while increased summer precipitation is cancelled by high summertime evaporation rates during these times (Abels et al. 2009). In their climate model, increased winter precipitation is due to elevated Mediterranean Sea temperatures and resulting cloud formation. Consequently, precession minima most likely resulted in a significant increase in the net water budget for the Teruel Basin. Following this reasoning, precession minima should be correlated with increased winter precipitation and lake-level highstands, and vice versa, in agreement with previous studies (Sierro et al. 2000; Kruiver et al. 2002; Abdul Aziz et al. 2003; Abels et al. 2009). Enhanced seasonality is suggested by increased color mottling in facies A to B in the meter-scale lithofacies cycles. Most probably the "wet" part (facies C, D, and E; see 4.3) of the lithofacies cycles and related groundwater table highstands in Prado relate to precession minima and association summer insolation maxima. The remaining uncertainties, however, lead us to consider both phase relations in the established astronomical tuning (Fig. 8).

6.3 Astronomical Tuning

Starting from the polarity reversals, the astronomical tuning of the Prado section is based on the correlation of successive lithofacies cycles to consecutive precession cycles (Fig. 8). The tuning of the "wet" part (see section 4.3) of the cycles to precession minima and 65° N summer insolation maxima is shown, as well as the alternative tuning following the opposite phase relation (Fig. 8). Solid lines in the Figure 8 are used when only one possible correlation exists, primarily due to the position and age of polarity reversals. Dotted lines instead indicate that the suggested correlation is the most probable, but an alternative correlation is possible with one precession cycle upwards or downwards. The resulting astronomical ages for the polarity reversals in the Prado section fall within the uncertainty intervals of the ages in the Monte dei Corvi section (Fig. 8). This implies that similar numbers of lithofacies cycles are present in the same intervals in both sections constrained by their independent, high-resolution magnetostratigraphy.

An uncertainty of one cycle exists in the tuning of cycles 14 to 18 which increases downwards to an uncertainty of two cycles. Therefore, the tuning is presented only for the "B" transect of the Prado section and not extended downwards to include the "A" transect. To solve the tuning of the cycles in transect "A" requires additional (magneto-)stratigraphic constraints, that have to come from locating of two cryptochrons of the long chron C5n.2n (Evans et al. 2007) in this lower part of the Prado section.

The astronomical tuning from cycles 13 to 30 is rather straightforward. The tuning in this interval depends on counting cycle 30 as a single or double cycle. If that cycle is considered as a double cycle, the tuning would shift one precession cycle downwards. In this case, the reversed chron found in cycles 23 and 24 still matches the ages for chron C5n.1r (Fig. 8; Hüsing et al. 2007). In addition, the match of the closely spaced cycles 18 and 19 with two closely spaced precession cycles at 10.05 Ma makes the presented tuning even more likely.

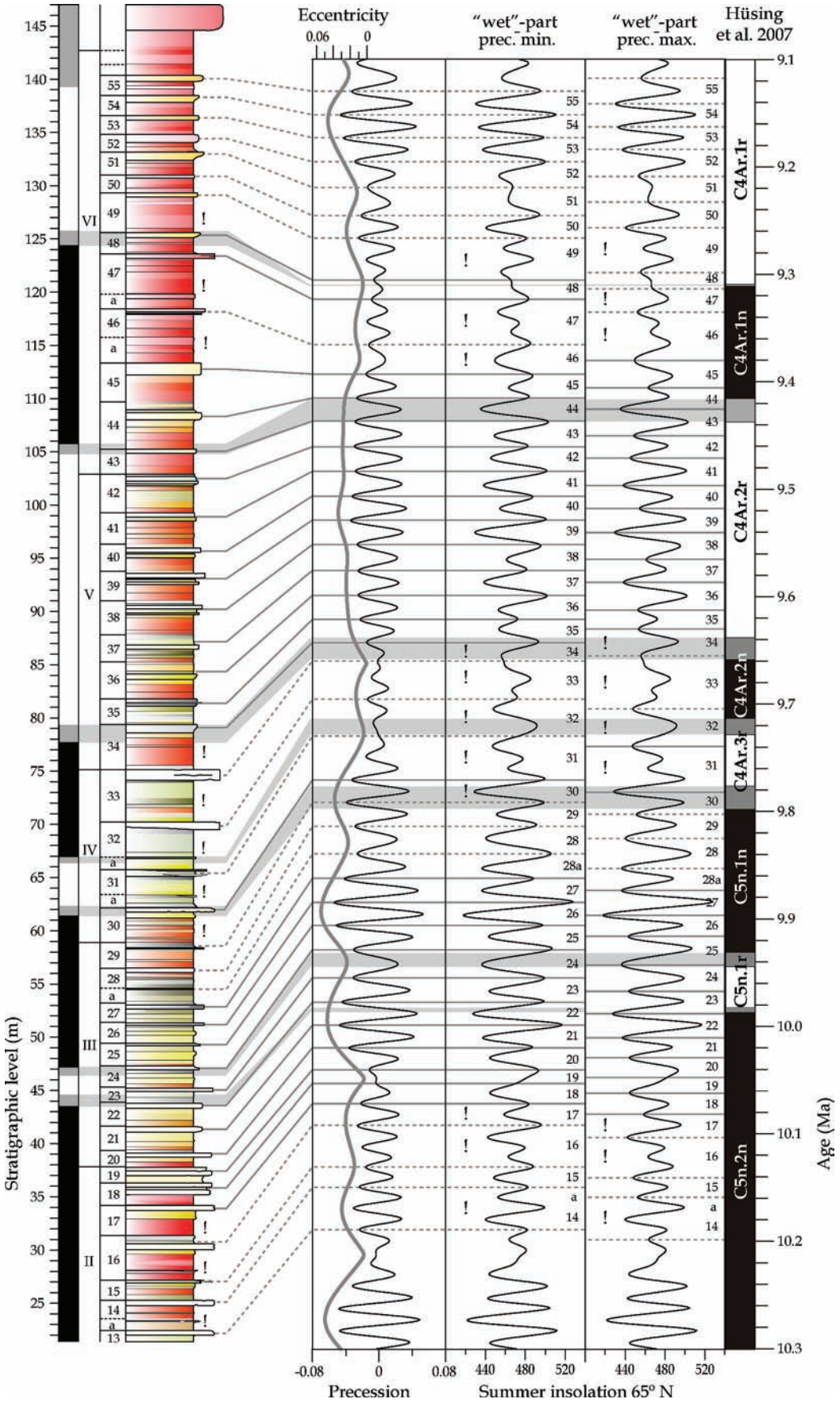
From cycle 30 to 34 and from cycle 45 to 49, the precise tuning of individual lithofacies cycles to precession is less uncertain. Here again, low eccentricity modulates precession, permitting an interplay between precession and obliquity. Nevertheless, cycles 34 to 45 are well constrained by the polarity reversals and the tuning seems to be correct.

The tuning reveals that the three intervals with thicker-than-average sedimentary cycles (cycles 16 and 17 in interval II, cycles 30 to 33 in interval IV, and cycles 46, 47, and 49) fall in 405 kyr eccentricity minima (Fig. 8). Such minima display reduced amplitudes of precession due to low eccentricity, thus allowing the 41 kyr obliquity cycle to interfere with precession. Consequently, the thickest cycles (14, 16, 17, 30 to 34, 46, 47, and 49; most including an "a" cycle and indicated by "!" in Fig. 8) may

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FIG. 7.— Declination, inclination, and interpreted polarity for the Prado magnetostratigraphy. To the left, a schematic column of the section is given, including stratigraphic position, and interval and cycles numbers. Black, white, and gray colors in the interpreted polarity column respectively indicate normal, reversed, and non-resolved polarity. The graphs to the right show the maximum average deviation (MAD), initial natural remanent magnetization (NRM), and the correlation of the interpreted polarity record at Prado to the ATNTS04 of Lourens et al. (2004).

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represent obliquity cycles, each comprising two precession cycles. This suggests that the "a" cycles should indeed be regarded as full precession cycles. Also, some of them may represent the longest or most extreme precession minima (or, oppositely, maxima).

The constructed astro-magnetostratigraphic age model for the Prado section thus provides a strong indication that the meter-scale lithofacies cycles are forced by climate variations dominantly driven by the *climatic* precession cycle. Additionally, the astronomical tuning reveals the imprint of periods of low 405 kyr eccentricity by reducing the amplitude of precession. In these periods, obliquity interferes with precession or even might dominate the climate forcing (Abdul Aziz et al. 2003; Hüsing et al. 2007). Astronomical forcing of sedimentation processes recorded in the Prado section is therefore regarded to be the most likely hypothesis underlying the three superimposed scales of lithofacies cyclicity.

6.4 Precession–Obliquity Interference Patterns

In Mediterranean deep-marine successions, distinct sedimentary patterns often show a close match with patterns in the summer insolation curve for 65° N (Lourens et al. 2001; Hüsing et al. 2007). The 65° N summer insolation curve is used here, because this target curve best reflects the Mediterranean marine sedimentary cycle patterns in the Neogene (Hilgen et al. 2000; Sierro et al. 2000; Lourens et al. 2001; Hilgen et al. 2003; Hüsing et al. 2007). In times of 405 kyr eccentricity minima the relative influence of obliquity on insolation becomes more prominent, and consequently obliquity can interfere with precession. The resulting distinct patterns in insolation curves are used to evaluate the match with sedimentary cycle patterns (Lourens et al. 2001). In the interval between 9.65 and 9.45 Ma in the Cascante section, the sedimentary record shows a distinct precession–obliquity interference pattern that approximately matches the P-0.5T (precession minus half obliquity) target curve (Abels et al. 2009), which is similar to the 65° N summer insolation curve. The evaluation of the tuning of the Prado cycles below suggests that a different target curve might be needed for sedimentary cycles in the Iberian Peninsula.

In the Prado section, alternating thick and thin "dry" parts of the cycle (i.e., facies A and B corresponding to groundwater-table lowstands) occur in the same time interval as in Cascante, confirming the presence of an obliquity–precession interference pattern in the stratigraphic column at Prado. However, the thick "dry" parts (cycles 34, 36, 38, 39, 41; Fig. 8) do not specifically match with extreme insolation minima, and thin "dry" parts do not match with relatively high insolation minima. On the contrary, a good pattern match is achieved when the opposite phase relation is regarded (Fig. 8), i.e., the "dry" part corresponding to precession minimum. At present, it is not possible to use this thickness mismatch with insolation as proof of an opposite phase relation because of the remaining uncertainties in the relation between local groundwater fluctuations and local climate and between local climate and precession, as discussed above. Moreover, it can be argued whether sedimentary cycles should be tuned to the 65° N summer insolation or a to different target curve.

6.5 Large-Scale Trends

Most large-scale paleoenvironmental shifts in the Prado section occur in intervals of 405 kyr eccentricity minima. The first shift is the change from well-drained mudflat environments in intervals I and II to more persistently waterlogged environments of intervals III and IV, at ~ 10.05 Ma. The reverse environmental change occurs in two steps. The first occurs at 9.65 Ma, from interval IV to V, in which well-drained environments re-enter the stratigraphy. This drying trend culminates in a second one occurring at 9.44 Ma, from interval V to VI, where well-drained mudflat to floodplain environments dominate. The relation between large-scale trends and 405 kyr eccentricity is suggestive, but the time span comprised by the sedimentary record in Prado is regarded to be too short to directly relate such paleoenvironmental shifts to eccentricity. Furthermore, while the 405 kyr minima at 9.35 Ma and 10.15 Ma are related to thick well-drained mudstones, the 405 kyr minimum at 9.7 Ma is related to thick waterlogged mudstones. This suggests that throughout the Prado section the sedimentary paleoenvironments would have reacted differently to long-eccentricity minima. It is then also plausible that other forcing mechanisms played a role, which may include tectonics, non-orbital climate variations, climate variations related to orbital cycles of longer periodicity (Abels et al. 2009), or a geomorphologic shift of the lacustrine environment.

7. DISCUSSION

Orbital climate forcing is shown to be the most plausible driver of paleoenvironmental change and therefore of local patterns in sedimentation in the Prado area by using an integrated stratigraphic and sedimentologic methodology. In addition, the cyclostratigraphic approach enables establishment of a high-resolution magneto-astrochronological framework and analysis of the meaning and architecture of facies variability at a temporal scale of 10^3 to 10^5 yr, which is usually hard to attain in ancient continental successions.

The Miocene facies organization on a meter scale in Prado is dominantly controlled by the climatic precession cycle, with a recognizable imprint by long, 405 kyr eccentricity and by obliquity on stratigraphic scales variable from meters to tens of meters. The influence of the precession cycle on the climate of the Iberian Peninsula and its signature in the geologic record has previously been identified in the late Cenozoic Spanish sedimentary record (Abdul Aziz et al. 2000; Sierro et al. 2000; Krijgsman et al. 2001; Sierro et al. 2001; Kruiver et al. 2002; Luzón et al. 2002; Abdul Aziz et al. 2004). Some of these studies also confirmed the concomitant role of obliquity and short and long eccentricity in coupling longer-term climate change to sedimentation (Krijgsman et al. 1994; Barberà et al. 1996; Sierro et al. 2000; Abdul Aziz et al. 2003; Abels et al. 2009).

Orbital forcing in the Prado section is demonstrated with the use of polarity-reversal ages that have been calculated by astronomical tuning of sapropel patterns in the Miocene Monte dei Corvi section, in northern Italy (Hüsing et al. 2007). Our reasoning thus highly depends on such astronomically tuned sedimentary successions that have a reliable magnetostratigraphy. The astronomical control of sedimentary cyclicity at Monte dei Corvi was confirmed by the astronomically tuned sapropels at Monte Gibliscemi (Hilgen et al. 2000; Hilgen et al. 2003). Further corroboration comes from the $^{40}Ar/^{39}Ar$ dating of biotites from two intercalated ash layers that indicate an average periodicity close to 20 kyr for basic cycles, of ~ 40 kyr for intermediate cycles, and of ~ 100 and ~ 400 kyr for the larger-scale cycles in the interval bracketed by the ash layers (Cleaveland et al. 2002; Hilgen et al. 2003). It must be noted that

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FIG. 8.—Astronomical calibration of the lithofacies cycles in Prado to precession and insolation target curves of Laskar et al. (2004), according to the two phase relations discussed in the text. To the left, a schematic log of the Prado section is shown, including the interpreted polarity results, and interval and cycle numbers. Cycle numbers are also indicated along the insolation curves. Thick cycles are indicated by "!" and occur during long, 405 kyr eccentricity minima. Gray-shaded bands indicate the correlation of polarity reversals and their respective uncertainty intervals in the Prado section to the astronomically calibrated reversals of the Monte dei Corvi in Italy (Hüsing et al. 2007).

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using astronomically calibrated ages for polarity reversals to construct an age model in an independent section is not equivalent to importing orbital cycle frequencies recognized in the section from which these ages were derived. This is because reversals of the Earth's magnetic field occur independently of orbital climate forcing. The results from the Prado section can rather be regarded as a corroboration of the constructed time scale and astronomical forcing hypothesis for the Monte dei Corvi section, similar to the corroboration based on comparing sapropel patterns of Monte dei Corvi and Monte Gibliscemi (Hilgen et al. 2003). The three cyclostratigraphic studies were carried out independently, even though Monte dei Corvi reversal ages were used as first-order age control for the Prado stratigraphy. The three sedimentary systems of these sections developed totally independently in separate marine environments and in the continental realm.

The cyclostratigraphy method applied in this study has a high potential for detecting orbital climate forcing in continental successions. Importing astronomically calibrated ages of reversal boundaries does not imply that the age model of the studied succession is biased towards astronomical forcing. Instead, if the astronomical time scale is correctly calibrated, then potentially the best possible detailed age model is used (Van Vugt et al. 1998; Kruiver et al. 2002; Abels et al. 2009). Also, the method avoids demonstration of astronomical forcing by comparing the lithofacies cyclicity with orbital cycle ratios derived from statistical analysis (Wilkinson et al. 2003; Meyers and Sageman 2007; Bailey and Smith 2008). Future statistical analysis of a sound, well-defined proxy record for water availability in the Prado paleoenvironment would, however, make the astronomical forcing hypothesis of this study stronger.

8. CONCLUSIONS

Astronomical climate forcing of facies organization is demonstrated in the continental Prado section as part of the upper Miocene fill of the endorheic, low-gradient southern Teruel Basin (northeast Spain). Detailed sedimentological logging coupled to stratigraphic analysis allowed the recognition of a meter-scale, composite lithofacies cycle. This cycle consists of an alternation of predominantly fine clastic and carbonate or carbonate-rich facies, and is interpreted in terms of relative variations through time of local groundwater level. The detailed magnetostratigraphic age model provided for the Prado section suggests that the meter-scale cyclicity is regulated by the climatic precession cycle. An astrochronology is established by correlating the individual meter-scale cycles to successive precession cycles. The resulting astronomical ages for polarity reversals at Prado are within the uncertainties of the time scale used. The age framework shows the superimposed imprint of long eccentricity and obliquity in modulating precession-scale cyclicity in three stratigraphic intervals. Uncertainties remain regarding the exact phase relation of groundwater to local climate and local climate to precession. Nevertheless, the astronomical tuning clearly reveals that orbital forcing was the prevailing forcing mechanism in the Prado mudflat to marginal lacustrine paleoenvironments at 10^3 yr to 10^5 yr time scales. This study shows that using astronomically calibrated ages for reversal boundaries is a useful method for demonstrating orbital forcing in nonmarine successions.

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